SHYAMPUR SIDDHESWARI MAHAVIDYALAYA
LIBRARY
Estd.- 1964

2021

COMPUTER SCIENCE — HONOURS

Paper : CC-10

(Microprocessor and its Applications)

Full Marks : 50

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1** and **2** and ***any three*** from the rest.

1. Answer ***any five*** questions from the following : 2×5

(a) What are the functions of $IO|\overline{M}$?

(b) How many register pairs are there in 8085?

(c) What is interrupt?

(d) How many flag registers are there? What is the necessity of flag register?

(e) What is ALE?

(f) What are the functions of PUSH and POP?

(g) Which type of data transfer technique is used if the speed of I/O devices do not match the speed of the microprocessor?

2. Write short notes on the following (***any two***) : 5×2

(a) Address bus-Data bus

(b) Interrupt-driven Data transfer Scheme

(c) Instruction cycle

(d) Programmable Peripheral Interface (PPI).

3. (a) How many machine cycles are required for the following instructions?

(i) MOV r_1, r_2;

(ii) MVI r, data;

(iii) LXI rp, data.

(b) Explain the requirement of Programme Counter and Stack Pointer. 6+4

4. (a) Show interfacing of memory and I/O devices, using decoder 74138. Explain how to determine memory addressing and I/O addressing locations for various zones.

(b) What is memory mapped I/O scheme? 8+2

5. (a) What are the differences between 'Burst mode' and 'Cycle stealing' techniques of DMA data transfer scheme?

(b) Compare and contrast between Synchronous and Asynchronous data transfer schemes. 6+4

6. (a) What are the various interrupt lines of 8085? Discuss their main features.

(b) Explain enabling, disabling and masking of interrupts. 5+5

7. (a) Interface a 4 kbyte EPROM with microprocessor 8085 in the memory range $FOOO_H$ and $FFFF_H$.

(b) What are the instructions used to access data from the ports in I/O mapped I/O method in microprocessor 8085? Explain with examples. 6+4

8. (a) What is the function of 8255?

(b) State briefly the function of 8279. 5+5

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